Assignment 4 DESIGN.pdf:

Description of Program:

This program provides a solution to the traveling salesman problem. It gets an input file from the user and creates a graph and finds the shortest hamiltonian path.

Files to be included in directory “asgn4” + pseudocode/structure:

- graph.c
  - implements the graph ADT.
- graph.h
  - specifies the interface to the graph ADT.
- path.c
  - implements the path ADT
- path.h
  - specifies the interface to the path ADT.
- stack.c
  - implements the stack ADT
- stack.h
  - specifies the interface to the stack ADT
- vertices.h
  - defines macros regarding vertices.
- tsp.c
  - contains main() and may contain any other functions necessary to complete the assignment.
    - -h : Help message. Prints the usage of the program
    - • -v : Verbose Printing. Prints each new optimal path as it is found
    - • -u : Uses an undirected graph
    - • -i infile : specifies the input file
    - • -o outfile : specifies the output file
- Makefile
  - CC = clang must be specified.
  - CFLAGS = -Wall -Wextra -Werror -Wpedantic must be specified.
  - make must build the Mathlib-test executable, as should make all and make mathlib-test
  - make clean must remove all files that are compiler generated.
  - make format should format all your source code, including the header files.
- README.md
  - In proper Markdown syntax.

    - Describe how to use your program and Makefile
    - Lists and explains any command-line options that program accepts.
    - Reports any false positives by scan-build
    - Notes down any known bugs or errors in this file as well for the graders.
- DESIGN.pdf
    - This document must be a proper PDF.
    - Describes design and design process for program with enough detail such that a sufficiently knowledgeable programmer would be able to replicate implementation.